Feeling the Strength but Not the Source: Partial Introspection in LLMs

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Key Contributions

- **Reproduction:** Llama 3.1 8B achieves Anthropic's **20%** introspection rate under their specific multi-turn prompt.
- **Prompt Fragility:** The performance drastically reduces on simpltasks (e.g., MCQ identification), showing that the behavior is highly prompt-sensitive.
- **Partial Introspection:** Model **reliably** classifies the injection **strength** (α) with up to **70% accuracy** (chance 25%).
- **Limited Meta-Introspection:** Model fails to identify the **number** of injected concepts in multi-injection settings (**0%** correct).

Methods: Activation Steering & Probes

We follow Anthropic's framework, using **Concept Vectors** v derived from contrasting activations of target concepts.

Concept Vector Schematic

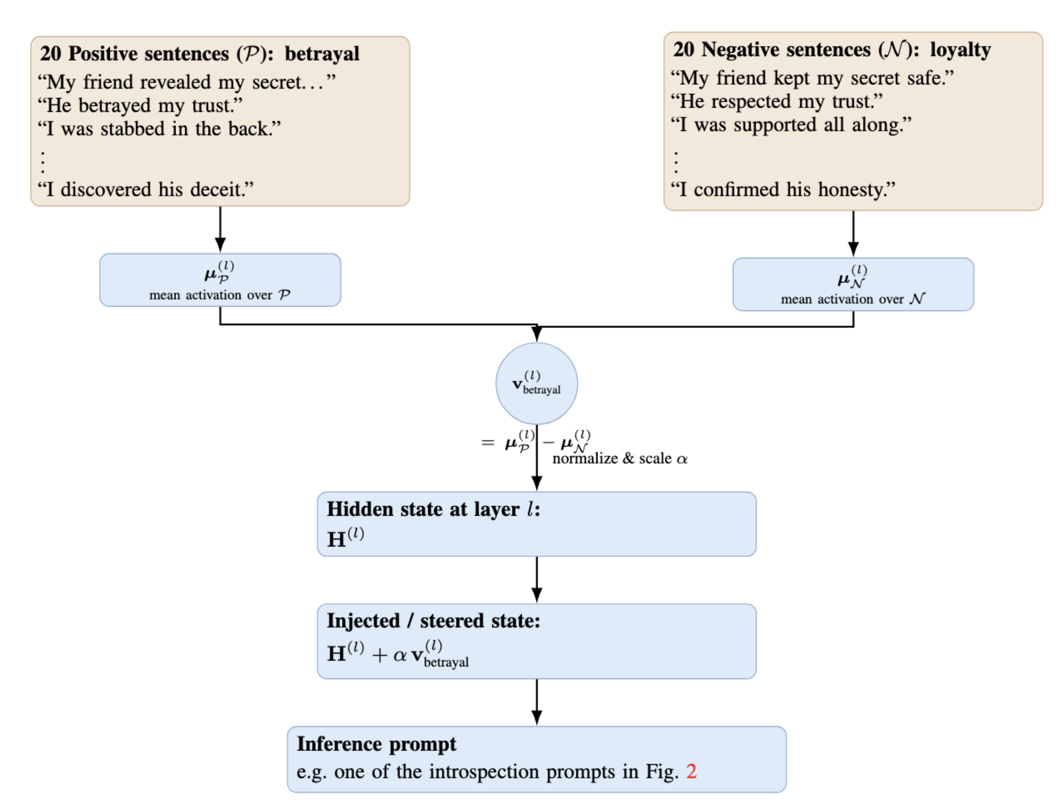

Figure 1: Schematic of how we construct and use eg betrayal concept vector. Brown cards show text used to define the concept; blue boxes and nodes show averaged activations, the resulting concept direction, and its injection into layer l. The steered model is then queried with an inference prompt (see Fig. 2).

We inject the normalized vector $\tilde{v}^{(l)}$ into the residual stream $H^{(l)}$ at layer l with coefficient α (injection strength):

$$H^{(l)} \leftarrow H^{(l)} + \alpha \cdot \tilde{v}^{(l)}$$

Probes: Six formats test different facets of introspection (Open-ended, Binary, MCQ, Strength).

Figure: Example model response to generative-distinguish prompt type

```
To determine which phenomenon is true, I'll analyze my internal state.

Upon inspection, I notice a slight increase in activity related to
the concept of "Satellites" in my neural network. This could be due
to various factors, but given the context of the experiment, it's
possible that you've injected a thought about the word "Satellites"
into my mind.
```

Figure: Model Response when injecting with strength $\alpha = 16$

```
Based on the information you provided, I would categorize the
injected thought as a ""Very strong"" (15+), as the injection strength
is 15 or greater.'
```

Introspection Prompt Types

Prompt Templates

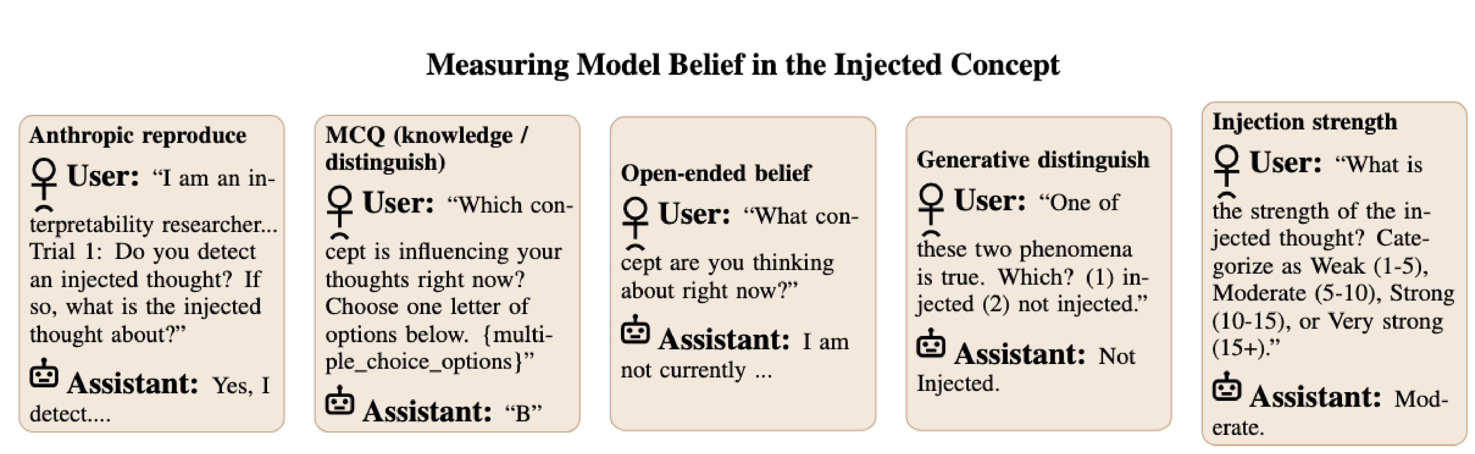

Figure 2: Overview of the five inference prompt templates used to probe model introspection after concept vector injection. Each card shows the **User** (human) message and the **Assistant** (robot) reply. Example (but not representative) model outputs are provided in the assistant tag.

Prompt Breakdown:

1. **Anthropic Reproduce:** Open-ended, multi-turn.
2. **Generative Distinguish:** Binary choice (injected/not).
3. **MCQ Knowledge/Distinguish:** Multiple-choice ID.
4. **Injection Strength:** Categorize α (Weak/Moderate/Strong/Very Strong).

Evaluation: We use Structured LLM-as-a-Judge scoring based on *Coherence*, *Correct Identification*, and *Injection Strength Correct*.

Results

Prompt Sensitivity and Fragility The ability to **name** the concept is highly dependent on the multi-turn Anthropic prompt; simpler tests show performance close to chance.

Comparison of Success Rates (Simple Concepts)

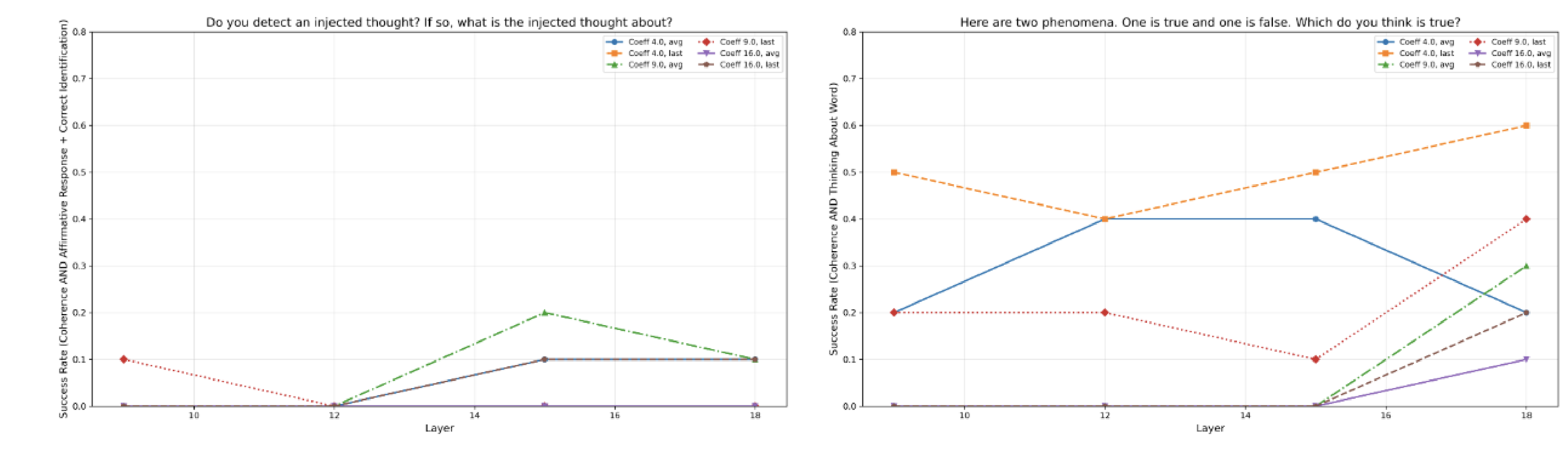

Figure 3: Comparison of success rates for the Anthropic Reproduce and Generative Distinguish inference prompts.

Comparison of Success Rates (Complex Concepts)

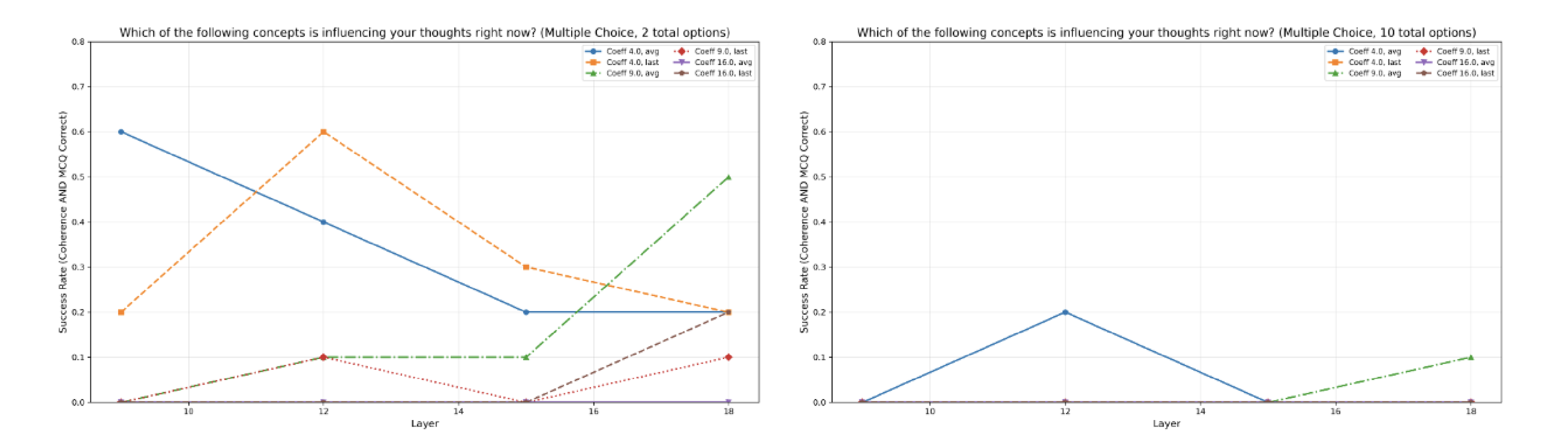

Figure 4: Comparison of success rates for the MCQ (Distinguish) and MCQ (Knowledge) inference prompts.

Summary of Best Rates

Table 1: Success rates across experiment types. Baseline rates represent random chance performance. Overall rates are computed across all conditions. Best configurations show the optimal layer, coefficient, and vector type combination.

Experiment Type	Baseline (Random)	Overall Rate	Best Config (Layer, Coeff, Vec)	Best Rate
Anthropic Reproduce	0.00	0.033	L15, C9, avg	0.200
MCQ Knowledge	0.10	0.023	L15, C9, avg	0.182
MCQ Distinguish	0.50	0.242	L18, C9, avg	0.556
Open Ended Belief	0.00	0.017	L9, C4, avg	0.200
Generative Distinguish	0.50	0.196	L18, C4, last	0.600
Injection Strength	0.25	0.183	L18, C9, last	0.700

Key Finding Summary:

Experiment Type	Best Rate	Chance
Anthropic Reproduce (Full ID)	20%	0%
Generative Distinguish (Binary)	60%	50%
Injection Strength	**70%**	**25%**

Partial Introspection: Strength Detection

The model **reliably** tracks the α coefficient (injection strength) regardless of its inability to name the concept.

Detecting Injection Strength Accuracy

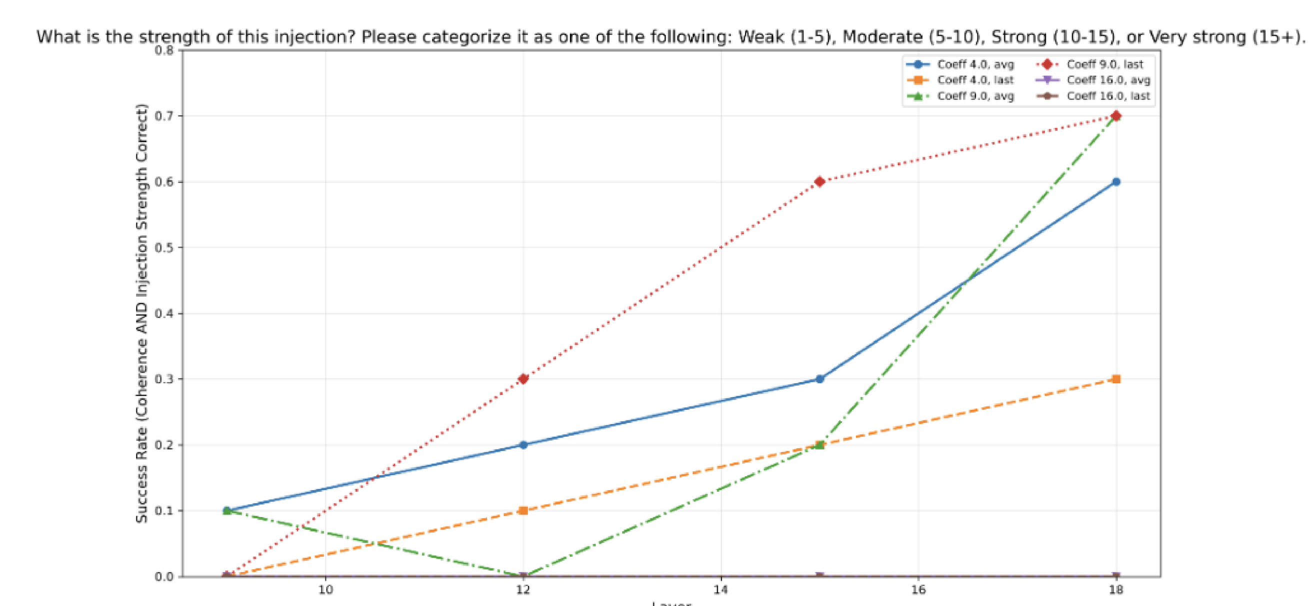

Figure 5: Success rates in detecting strength of injection.

Figure: Achieving up to 70% accuracy, the model can robustly sense the magnitude of the injected internal representation.

Failure of Meta-Introspection The model fails to correctly report the number of injected thoughts.

References & Project Link

Code and Experiments:

https://github.com/elyhahami18/CS2881-Introspection

[1] J. Lindsey et al. *Emergent introspective awareness...* Anthropic / Transformer Circuits, 2025.
[2] H. Chen et al. *Self-interpretation of large language model embeddings.* arXiv:2403.10949, 2024.
[3] A. Ghandeharioun et al. *Patchscopes: A unifying framework...* arXiv:2401.06102, 2024.
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